

To **fully back up and restore Grafana** (dashboards, data sources, users, alerts, settings) so you can recover from a crash or re-install, you mainly need to preserve **three things**.

1 Back up the Grafana data directory (MOST IMPORTANT)

Grafana stores *almost everything* in its data directory.

Default locations

- Linux package install:
`/var/lib/grafana/`
- Docker:
Whatever volume is mapped to `/var/lib/grafana`
- Custom install:
Check `paths.data` in `grafana.ini`

What's inside

- `grafana.db` → **SQLite database** (dashboards, users, alerts, data sources)
- `plugins/` → Installed plugins [**DO NOT BACK THESE UP, they are huge and not custom**]
- `png/`, `csv/` → Rendered/exported files (if used)

Backup command (example)

```
sudo systemctl stop grafana-server
tar czvf grafana-data-backup.tar.gz /var/lib/grafana
sudo systemctl start grafana-server
```

- ♦ Stopping Grafana avoids database corruption during backup.
-

2 Back up configuration files

These define how Grafana runs and connects to things.

Files to save

```
/etc/grafana/grafana.ini
/etc/grafana/provisioning/
```

Provisioning includes:

- Data sources
- Dashboards
- Alerting rules (if provisioned)
- Notifiers

Backup example

```
tar czvf grafana-config-backup.tar.gz /etc/grafana
```

(Optional but Recommended) Export dashboards as JSON

This gives you **human-readable, versionable** backups.

Manual export

- Dashboard →  Settings → **JSON Model** → Save

Automated export (API)

Grafana has a REST API to export dashboards:

```
GET /api/dashboards/uid/:uid
```

Useful if you want:

- Git-based backups
 - CI/CD style restores
-

Restore procedure (after crash or reinstall)

1. Install the **same or newer** version of **Grafana**

2. Stop Grafana

3. Restore files:

```
sudo tar xzvf grafana-data-backup.tar.gz -C /  
sudo tar xzvf grafana-config-backup.tar.gz -C /
```

4. Fix ownership:

```
sudo chown -R grafana:grafana /var/lib/grafana /etc/grafana
```

5. Start Grafana:

```
sudo systemctl start grafana-server
```

Your dashboards, users, alerts, and data sources should reappear exactly as before.

Minimal “bulletproof” backup checklist

If you only do **three things**, do these:

-  /var/lib/grafana/
-  /etc/grafana/grafana.ini
-  /etc/grafana/provisioning/

That’s enough for a **100% restore** in almost all setups.



Full Grafana Backup Script

Save this as:

```
/usr/local/bin/grafana-full-backup.sh
```

Make it executable:

```
sudo chmod +x /usr/local/bin/grafana-full-backup.sh
```

```
#!/bin/bash

# =====
# Grafana Personal Backup Script
# =====

# Directories
BACKUP_DIR="/home/tom/grafana_backups" # change if needed
DATA_FILE="/var/lib/grafana/grafana.db"
CONFIG_DIR="/etc/grafana"

# Grafana service
SERVICE_NAME="grafana-server"

# Retention in days
RETENTION_DAYS=14

# Timestamp
DATE=$(date +%Y-%m-%d_%H-%M-%S)
BACKUP_FILE="$BACKUP_DIR/grafana-personal-backup-$DATE.tar.gz"
LOG_FILE="$BACKUP_DIR/backup.log"

# Ensure backup directory exists
mkdir -p "$BACKUP_DIR"
```

```
echo "-----" >> "$LOG_FILE"
echo "Personal backup started at $(date)" >> "$LOG_FILE"

# Stop Grafana safely
echo "Stopping Grafana service..." >> "$LOG_FILE"
sudo systemctl stop $SERVICE_NAME

# Create archive of DB and config only
echo "Creating archive..." >> "$LOG_FILE"
sudo tar -czpf "$BACKUP_FILE" "$DATA_FILE" "$CONFIG_DIR"

# Start Grafana
echo "Starting Grafana service..." >> "$LOG_FILE"
sudo systemctl start $SERVICE_NAME

# Remove old backups
echo "Removing backups older than $RETENTION_DAYS days..." >> "$LOG_FILE"
find "$BACKUP_DIR" -name "grafana-personal-backup-*.tar.gz" -mtime +$RETENTION_DAYS
-delete

echo "Backup completed at $(date)" >> "$LOG_FILE"
echo "Backup file: $BACKUP_FILE" >> "$LOG_FILE"
```



Automate with Cron

Edit root's crontab:

```
sudo crontab -e
```

Add a line to run nightly at 2AM:

```
0 2 * * * /usr/local/bin/grafana-full-backup.sh
```



Restore Procedure

If something goes wrong, restore **both directories**:

```
# Stop Grafana
sudo systemctl stop grafana-server

# Extract backup
sudo tar -xzipf grafana-full-backup-YYYY-MM-DD_HH-MM-SS.tar.gz -C /

# Fix ownership
sudo chown -R grafana:grafana /var/lib/grafana /etc/grafana

# Restart Grafana
sudo systemctl start grafana-server
```

Your dashboards, users, alerts, data sources, provisioning, and custom settings are fully restored.